



Tuka Curious Minds

MASTER ROBO WARS DESIGN

CONCEPT SUBMISSION

Robot Name: TURBO Ceros-T

Weight Classification: 5.kg Max Limit

Footprint Matrix: 420mm (Width) x 420mm (Length) x 55mm (Sleek Casing Height)

MODULE C: FABRICATION CONSIDERATIONS & CONSTRUCTION BRIEF

Target Chassis Footprint: 420mm x 420mm x 55mm

Primary Workshop Tools Required: Handheld Angle Grinder (1mm Inox Discs + 80-Grit Flap Disc), Hand Drill or Drill Press, Center Punch, Basic Hex Keys.

1. Fabrication Considerations (Material & Tool Strategy)

When working with a mix of heavy steel, soft aluminum, and pre-drilled structural kit channels using basic workshop hand tools, the fabrication team must follow these four core guidelines to prevent structural failures:

1.1 Steel vs. Aluminum Grinder Techniques

- **The 2mm Mild Steel Armor:** Use a **1mm ultra-thin Inox cutting disc** on your angle grinder. Thin discs cut through steel quickly with minimal heat buildup, preventing the metal edges from warping. Always clamp the steel offcuts firmly to a workbench before cutting.
- **The 1.5mm Aluminum Gate Sheets:** Aluminum is soft and melts easily. A regular steel cutting disc will clog up with melted aluminum and can shatter violently. Use a dedicated **aluminum-rated cutting disc** or score your cut lines lightly, then clamp the sheet between two pieces of wood to cleanly bend and snap it along the score line.
- **The Sharp "Razor" Edge Face:** To win the ground-clearance war against opponent wedges, use an **80-grit flap disc** on the angle grinder. Grind the bottom edge of the red front steel wedge downward at a 15-degree angle until it sits microscopically flat against the arena floor.

1.2 Center-Punching & Dual-Stage Drilling (Anti-Binding Rule)

- **The Problem:** If you try to drill holes through your raw steel armor plates by eye, the drill bit will skate across the metal. If a hole is off by even 1 millimeter, the bolts will not line up with the pre-drilled 8mm goBILDA frame channels, causing the chassis to bind.
- **The Fix:** Coat the steel plates in black marker or dark primer, scratch your exact grid intersections onto it, and strike the center cross lines firmly with a hardened **center punch** to leave a deep physical dimple.
- **The Two-Step Drill:** First, drill a pilot hole using a tiny **2mm drill bit**. Second, open the hole using a larger **4.5mm HSS (High-Speed Steel) bit**. The extra 0.5mm clearance gives you structural wiggle room, allowing every single M4 screw to drop smoothly into the internal aluminum frame threads.

- **Lubrication:** Dip the drill bit into motor oil or soapy water every three holes to keep the metal cool and prevent the cutting edges from melting.

2. Step-by-Step Construction Brief (Assembly Sequence)

Follow this chronological build order to systematically layer the chassis skeleton, internal components, and outer armor casing into a single tournament-ready machine:

[STEP 1: INNER C-FRAME] → [STEP 2: MOTORS & WHEELS] → [STEP 3: SAWS TOWER]
|
[STEP 6: TOP COVER SEALS] ← [STEP 5: CORE ELECTRONICS] ← [STEP 4: STEEL WEDGE]

Step 1: Assembling the Internal Grid Skeleton

1. Layout the four outer **heavy C-channels** facing inward to form your square **420mm x 420mm boundary tray**.
2. Slide the **aluminum U-channel spines** from your kit inside to act as front-to-back cross-bracing structural members.
3. Secure every frame joint using high-tensile M4 hex capscrews tightened directly into **nylon-insert locknuts**. Never use standard nuts; the vibrations from the arena floor and weapons will shake them loose within 30 seconds.

Step 2: Mounting the Propulsion Units & Wheels

1. Bolt **REV Core Hex Motors #1 and #2** into the rear-left and rear-right internal U-channel tracks using their multi-pattern face mounting plates.
2. Align your two heavy **rear traction wheels** onto the output axles. Pass the wheel shafts through internal bearing pillow blocks to ensure all lateral impact shocks are absorbed by the frame structure rather than the motor box gearboxes.
3. Slide your **two passive front Omni Wheels** straight into the front-left and front-right corner corridors of the C-channel frame. Pin them in place on non-driven passive steel shafts.

Step 3: Installing the Forward Overhung Weapon Dead-Shaft

1. Clamp your custom **triangular steel tower block** (fabricated from your 2mm steel scraps) onto the absolute front-center edge of the frame U-channels, allowing it to overhang forward past the bumper boundary line.
2. Run your stationary **12mm hardened steel dead-shaft bolt** horizontally through the peak of the tower. Clamp it down tightly with shaft collars so it remains completely rigid and **does not rotate**.
3. Press high-speed ball bearings into the center bores of your **two 200mm circular saw discs**. Slide them onto the 12mm dead shaft so they spin symmetrically on the left and right faces of the tower.
4. Mount the **6,000 RPM high-speed motor** in the middle bay and loop the **15mm wide HTD 5mm rubber timing belt** forward to the center saw pulley hub.

Step 4: Mounting the Front Wedge & Solenoid Windows

1. Take your grinder-cut **Red & Black 2mm steel nose plate** and align it over the front ramp frame rails. Ensure the pre-cut **12mm wide vertical saw slits** clear the blades perfectly.

2. Position the **two 12V linear solenoids** inside the left and right front structural corners. Push their silver actuator rods forward through the pre-cut **20mm x 40mm horizontal armor windows**.
3. Attach the small external aluminum trapping flippers to the tips of the solenoid rods on the floor outside the wedge. Bolt the front wedge plate down to the frame chassis skeleton.

Step 5: Wiring the Electrical Core Bay

1. Secure the **REV Control Hub MCU** flat against the rear-center bay floor using dense foam padding strips to isolate the circuit boards from intense weapon vibration harmonics.
2. Slide your single **12V NiMH battery brick** vertically into its custom-folded aluminum safety cage next to the hub.
3. Route all electrical wires, signal lines, and battery leads cleanly inside the hollow gutters of the inward-facing **steel C-channels**, away from the moving weapon belt and tires. Connect the signal leads from Digital I/O Port 0 to your **12V Isolated Optical Relay Switchboard Module**.

Step 6: Final Torsion Sealing & E-Stop Compliance

1. Fasten your continuous flat **1.5mm aluminum gate-sheet belly pan** across the absolute bottom of the chassis rails to seal the robot's underside floor.
2. Mount the high-visibility, red **Emergency Stop Button** flush through the top casing deck plate cutout.
3. Drop the sleek, low-profile **white aluminum upper casing deck** flat across the top of the 55mm tall framework structure. Bolt it down securely into the channel grid to turn the machine into a rigid structural torsion box, ready for tech inspection.